Deven Desai

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SUMMARY

Software Engineering Master's student and Stanford Hackathon winner with deep systems programming experience, C++ and Go fluency, and hands-on expertise building low-latency, concurrent distributed engines.

PROJECTS

Low Latency Search Engine | High-Performance Storage & Retrieval Engine GitHub

- Queried and analyzed massive structured datasets in real time by engineering a custom high-performance indexing interface as a solo developer using C++ and Java.
- Decreased query latency by 35 percent by developing low-level memory access routines, custom indexing data structures, and cache-friendly algorithms as measured by benchmark traces.
- Optimized processing throughput by 20 percent by profiling memory allocations, tuning concurrency primitives, and eliminating critical execution bottlenecks.

AgenticFlow | Concurrent Distributed Systems Engine GitHub

- Automated schema migrations and data transformations across cloud systems by architecting a high-throughput distributed execution service using Go and Python.
- Increased service execution throughput by 40 percent by designing concurrent execution loops, lock-free communication patterns, and low-overhead gRPC interfaces.
- Ensured strict system reliability and data consistency across execution boundaries by engineering an automated differential privacy and verification layer.

OmniTrack AI | High-Throughput Distributed Data Processing Engine GitHub

- Processed unstructured live streaming data natively in cloud infrastructure by engineering a zero-to-one distributed data pipeline using Python and C++.
- Achieved sub-100ms vector retrieval latency by independently optimizing Qdrant storage indexing algorithms for compute-intensive, concurrent requests.
- Enhanced system stability and output correctness by 30 percent by building continuous execution monitoring and automated testing suites.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

Systems & Low-Level Engineering: Distributed Storage Systems, Concurrency & Multithreading, Memory Optimization, Cache Efficiency
Languages: C++, C, Go, Python, Rust Concepts, Java, SQL, TypeScript
Distributed Architecture: Distributed Systems, gRPC Microservices, RESTful APIs, High-Throughput Pipelines, Data Replication
Developer Tools & Practices: Git, Docker, Performance Profiling, System Benchmarking, CI CD, Automated Testing, Code Reviews

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.